

UPSSITECH — Systèmes Robotiques Interactifs (SRI)

4th Year Engineering — Internship Report

LMI-Based Multirate Steady-State Kalman Filter Design: Comparison of Offline and Recursive Gains

Student

Léo Ahmed Mushtaq
4th Year, UPSSITECH SRI
Université Paul Sabatier — Toulouse
III
Academic year 2025–2026

Supervisor

Prof. Hiroshi Okajima
Associate Professor
Kumamoto University, Japan
Laboratory of Control Engineering

Host Institution

Kumamoto University
Graduate School of Science and Technology
Kumamoto, Japan

Internship period: May – August 2026 (3.5 months)

Résumé détaillé

Ce stage de 3 mois et demi se déroule au sein du laboratoire du Professeur Hiroshi Okajima à l'Université de Kumamoto (Japon), dans le cadre de la 4^{ème} année d'ingénieur de la filière Systèmes Robotiques et Interactifs (SRI) de l'UPSSITECH (Toulouse).

Contexte et problématique. Dans de nombreux systèmes de contrôle modernes, plusieurs capteurs opèrent à des fréquences d'échantillonnage différentes. Un exemple typique est la navigation automobile : le GPS fournit une position à 1 Hz, tandis que le capteur de vitesse de roue transmet des données à 10 Hz. Ce contexte, appelé système *multirate* (multi-cadence), génère une dynamique de mesure périodique dans le temps qui invalide les hypothèses du filtre de Kalman classique. Plus précisément, lorsqu'on empile les mesures sur une période complète N (reformulation cyclique), la matrice de covariance de bruit de mesure $\check{R}$ devient semi-définie positive — non définie positive — ce qui empêche l'utilisation directe de l'équation de Riccati algébrique discrète (DARE). C'est la contribution centrale de l'article de référence du stage : H. Okajima, *LMI Optimization-Based Multirate Steady-State Kalman Filter Design*, IEEE Access, 2025.

Approche LMI. Pour contourner le problème de la singularité de $\check{R}$, le stage s'appuie sur une formulation par Inégalités Matricielles Linéaires (LMI). La reformulation cyclique transforme le système périodique en un système linéaire invariant dans le temps (LTI) augmenté de dimension $Nn \times Nn$. L'équation de Lyapunov associée, contrairement à la DARE, ne nécessite pas l'inversion de $\check{R}$ et reste valide même en cas de singularité. En introduisant le changement de variable $\check{Y} = -\check{X}\check{L}$ (linéarisation du produit), le problème de design se réduit à un programme semi-défini (SDP) convexe : minimiser $\text{trace}(\check{W})$ sous contraintes LMI, où $\check{W} \succeq \check{X}^{-1}$ est une borne supérieure sur la covariance d'erreur $\check{P}_e$.

Objectif du stage — les trois jalons. Le travail se décompose en trois jalons successifs. Le **Jalon 1** consiste à reproduire fidèlement l'exemple numérique du papier (GPS 1 Hz + roue 10 Hz, $N = 10$, $n = 3$) : vérification de la construction cyclique, rang de $\check{R}$, gains périodiques L_k , stabilité ($\max|\lambda| < 1$) et performances RMSE. Le **Jalon 2** consiste à implémenter la récursion de Kalman *time-varying* : à chaque instant k , la covariance de prédiction $P(k|k-1) = AP(k-1|k-1)A^\top + Q$ est mise à jour par le gain actif $K(k) = P_{\text{pred}}C_k^\top(C_kP_{\text{pred}}C_k^\top + R_k)^{-1}$, en n'inversant que le bloc actif non singulier. Le gain récursif est ensuite converti en forme prédictive via $L_{\text{rec}}(k) = A \cdot K(k)$ pour être comparable aux gains LMI. Le **Jalon 3** analyse l'écart éventuel entre $L_{\text{rec}}(k)$ en régime permanent et L_k LMI, et l'interprète en termes de conservativité : la LMI minimise une *borne supérieure* sur la covariance, pas la covariance elle-même ; lorsque $\check{R}$ est semi-définie, les deux gains peuvent différer légèrement.

Travaux réalisés à mi-stage. Les deux premières semaines ont été consacrées à la

consolidation des fondements théoriques : représentation d'état, commandabilité et observabilité, placement de pôles, commande LQR, stabilité de Lyapunov en temps discret, et norme H_∞ . La syntaxe native de la Robust Control Toolbox MATLAB (`setlmi`, `lmivar`, `lmiterm`, `mincx`) a été maîtrisée sur des exemples académiques (pendule inversé) avant d'aborder le cas multirate.

Compétences mobilisées. Ce stage articule des compétences d'automatique avancée (théorie des systèmes périodiques, optimisation convexe, dualité LQR/filtre de Kalman), de programmation MATLAB scientifique (construction de matrices structurées, solveurs SDP), et de rigueur scientifique (validation numérique, analyse d'erreur, rédaction académique).

Abstract

This report presents the work carried out during a 3.5-month research internship at Kumamoto University (Japan) under the supervision of Professor Hiroshi Okajima. The project focuses on the design of optimal state estimators for multirate systems, where sensors operate at different sampling rates.

The central contribution of the associated paper (Okajima, IEEE Access, 2025) is a Linear Matrix Inequality (LMI) framework that resolves a fundamental limitation of classical DARE-based Kalman filtering: when the cyclic measurement noise covariance $\tilde{R}$ becomes positive semidefinite due to the multirate structure, standard solvers fail. The LMI approach naturally handles this singularity through a dual LQR formulation.

The internship is organized around three milestones: (1) reproduction of the automotive navigation numerical example (GPS at 1 Hz, wheel speed at 10 Hz, $N = 10$, $n = 3$); (2) implementation of a time-varying Kalman recursion and conversion to predictor form via $L = AK$; (3) quantitative comparison of the offline LMI gains and the recursive steady-state gains, with interpretation of any residual gap in terms of the LMI's conservativeness.

Keywords: Multirate Kalman filter, LMI optimization, cyclic reformulation, semidefinite covariance, periodic systems, sensor fusion.

Contents

Résumé détaillé (français)	1
Abstract (English)	3
1 Introduction	5
2 Context and Objectives	5
2.1 Host institution — Kumamoto University	5
2.2 The multirate estimation problem	6
2.3 Objectives of the internship	7
3 Theoretical Background and State of the Art	8
3.1 State-space representation and observability	8
3.2 Standard Kalman filter and its limitations	8
3.3 The semidefinite challenge in multirate systems	8
3.4 LMI optimization and semidefinite programming	9
4 Technical Contribution	9
4.1 Cyclic reformulation	9
4.2 LMI-based filter design	10
4.3 Multi-objective extensions	12
4.4 Milestone 2 — Time-varying Kalman recursion	12
4.5 Milestone 3 — Gain comparison and gap analysis	13
5 Results and Analysis	13
5.1 Milestone 1 — Reproduction results	13
5.2 Milestone 2 — Recursive gain convergence	15
5.3 Milestone 3 — Gain comparison	16
6 Assessment	18
6.1 Scientific and technical assessment	18
6.2 Personal assessment	19
Conclusion	20
A Complete System Matrices	22
B MATLAB Implementation Notes	22

1 Introduction

Modern engineering systems increasingly rely on heterogeneous sensor networks where different measurement devices operate at incompatible sampling rates. In autonomous vehicle navigation, GPS receivers typically update at 1–10 Hz while inertial measurement units and wheel speed encoders run at 50–200 Hz. Similar challenges arise in chemical process control, power systems monitoring, and robotic perception pipelines.

This four-month research internship takes place at Kumamoto University (Japan) in the laboratory of Associate Professor Hiroshi Okajima, a specialist in control engineering and LMI-based optimization with over 20 years of research experience. The host laboratory works on state estimation, model error compensation, and multirate control systems, with a strong emphasis on open-access publications and reproducible MATLAB implementations.

The central task of the internship is to study, reproduce, and extend the results of the paper [1], which proposes an LMI-based framework for designing optimal steady-state Kalman filters in multirate systems. The key scientific question is: *do the offline periodic gains computed by LMI optimization coincide with the steady-state gains obtained by running the standard time-varying Kalman recursion?*

This report is structured as follows. Section 2 describes the laboratory, the application context, and the project objectives. Section 3 reviews the theoretical background: state-space representations, Kalman filtering, and LMI optimization. Section 4 details the three milestones: reproduction, recursive gain computation, and comparison. Section 5 presents the numerical results and their analysis. Section 6 provides the scientific and personal assessment of the internship.

2 Context and Objectives

2.1 Host institution — Kumamoto University

Kumamoto University is a national research university located in Kumamoto City, Japan. Its Graduate School of Science and Technology hosts several control engineering research groups. Professor Hiroshi Okajima’s laboratory focuses on:

- Multirate control and estimation systems;
- Model error compensator (MEC) design;
- LMI-based robust controller synthesis;
- Open-access MATLAB implementations for reproducibility.

The laboratory methodology emphasizes rigorous mathematical foundations paired with fully reproducible numerical experiments, a philosophy that directly shapes this internship.

2.2 The multirate estimation problem

Consider a discrete-time linear time-invariant (LTI) system:

$$x(k+1) = Ax(k) + Bu(k) + Q^{1/2}d_w(k), \quad (1)$$

$$y(k) = S_k Cx(k) + S_k R^{1/2}d_v(k), \quad (2)$$

where $x(k) \in \mathbb{R}^n$ is the state, $u(k) \in \mathbb{R}^p$ the control input, and $d_w(k), d_v(k)$ are normalized white noise processes with zero mean and unit covariance. The matrices $Q = Q^{1/2}(Q^{1/2})^\top \succeq 0$ and $R \succ 0$ are the process and measurement noise covariances respectively.

The matrix $S_k = \text{diag}(s_{k,1}, \dots, s_{k,q}) \in \mathbb{R}^{q \times q}$ is a *periodic selection matrix*: $s_{k,i} = 1$ if sensor i is active at time k , and 0 otherwise. When all sensor sampling rates have rational ratios, there exists a period N such that $S_{k+N} = S_k$ for all $k \geq 0$.

Application example. The running example throughout this report is an automotive navigation system fusing:

- A **GPS receiver** providing position measurements at 1 Hz;
- A **wheel speed encoder** providing velocity measurements at 10 Hz.

The state is $x(k) = [p(k), v(k), a(k)]^\top$ (position, velocity, acceleration), $n = 3$, $q = 2$, $N = 10$, sampling period $\Delta t = 0.1$ s. The system matrices are:

$$A = \begin{pmatrix} 1 & 0.1 & 0.005 \\ 0 & 1 & 0.1 \\ 0 & 0 & 0.8 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad (3)$$

with noise covariances $Q = \text{diag}(0.01, 0.1, 0.5)$ and $R = \text{diag}(1.0, 0.1)$.

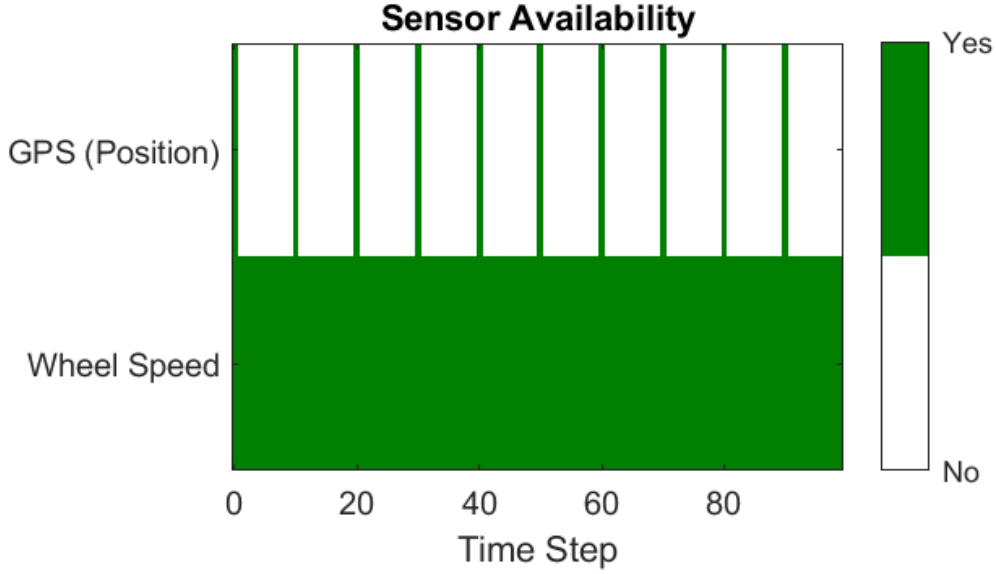


Figure 1: Measurement availability pattern for the automotive navigation system. GPS provides position updates at 1 Hz (every 10 steps), while the wheel speed encoder provides velocity measurements at 10 Hz (every step). The selection matrix alternates between $S_0 = \text{diag}(1, 1)$ (both sensors active) and $S_k = \text{diag}(0, 1)$ (wheel speed only) for $k = 1, \dots, 9$.

2.3 Objectives of the internship

The internship is structured around three progressive milestones defined jointly with Professor Okajima:

1. **Milestone 1 — Reproduction:** Reproduce the numerical results of [1] using the provided MATLAB code. Verify the cyclic construction, the semidefinite structure of $\tilde{R}$, the periodic gains L_k , and the reported stability and RMSE metrics.
2. **Milestone 2 — Recursive gain:** For the same periodic system, implement the standard time-varying Kalman recursion, updating the gain at each step k from the predicted covariance $P(k|k-1)$. Plot the gain trajectory and observe convergence to a periodic steady state.
3. **Milestone 3 — Comparison:** Convert both gains to the same predictor form ($L = AK$) and compare them after the transient. Quantify and interpret any residual gap between the LMI gain and the recursive steady-state gain.

3 Theoretical Background and State of the Art

3.1 State-space representation and observability

The state-space framework of equations (1)–(2) was systematically studied at the beginning of the internship using academic examples (third-order systems, inverted pendulum). A key observation is that pole-zero cancellation in the transfer function corresponds to an unobservable mode in the state-space representation: if a pole of A is also a zero of the system, the corresponding internal state is hidden from the output $y(t)$.

Observability is verified by checking the rank of the observability matrix $\mathcal{O} = [C^\top, (CA)^\top, \dots, (CA^{n-1})^\top]^\top$. Full rank ($\text{rank}(\mathcal{O}) = n$) is a necessary condition for any state estimator to recover all states from the measured outputs. For the cyclic system studied in this report, $\text{rank}(\mathcal{O}_{\text{cyc}}) = 30 = Nn$, confirming that the augmented system is fully observable despite the intermittent GPS signal.

3.2 Standard Kalman filter and its limitations

The classical discrete-time Kalman filter (predictor form) computes, at each step k :

$$P(k|k-1) = AP(k-1|k-1)A^\top + Q, \quad (4)$$

$$K(k) = P(k|k-1)C^\top (CP(k|k-1)C^\top + R)^{-1}, \quad (5)$$

$$P(k|k) = (I - K(k)C)P(k|k-1), \quad (6)$$

$$\hat{x}(k+1|k) = A\hat{x}(k|k-1) + Bu(k) + L(k)(y(k) - C\hat{x}(k|k-1)), \quad (7)$$

where $L(k) = AK(k)$ is the predictor-form gain. Under stationarity conditions, the recursion converges to a steady state given by the Discrete Algebraic Riccati Equation (DARE):

$$P = APA^\top + Q - APC^\top (CPC^\top + R)^{-1}CPA^\top. \quad (8)$$

The DARE requires $R \succ 0$ to compute the inverse $(CPC^\top + R)^{-1}$. This is the fundamental limitation in the multirate setting.

3.3 The semidefinite challenge in multirate systems

After cyclic reformulation (Section 4.1), the measurement noise covariance becomes:

$$\check{R} = \text{diag}(S_0RS_0^\top, S_1RS_1^\top, \dots, S_{N-1}RS_{N-1}^\top). \quad (9)$$

For the automotive example, $\check{R} \in \mathbb{R}^{20 \times 20}$ has rank 11 (out of 20): the nine blocks corresponding to $k = 1, \dots, 9$ (GPS inactive, $S_k = \text{diag}(0, 1)$) have rank 1 instead of 2.

Consequently, $\tilde{R}$ is positive *semidefinite*, not positive definite, and the DARE cannot be applied directly.

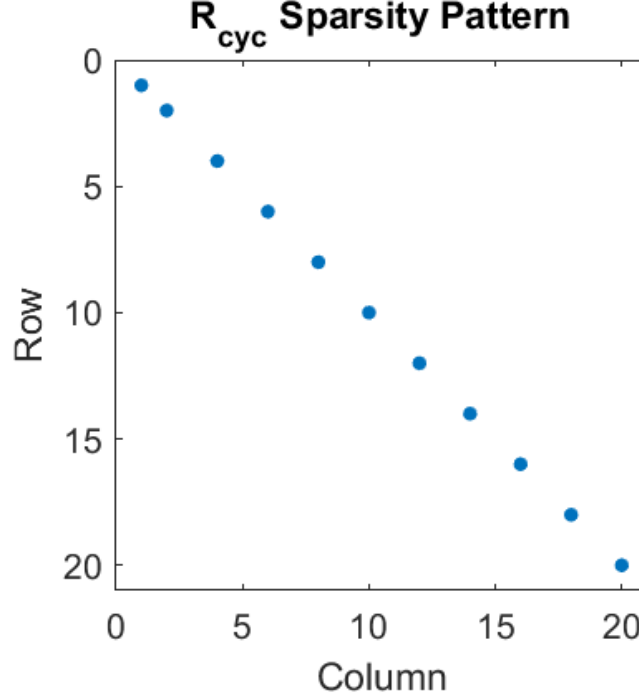


Figure 2: Structure of the cyclic measurement noise covariance $\tilde{R} \in \mathbb{R}^{20 \times 20}$. The matrix has rank 11 (out of 20): only the diagonal entry corresponding to wheel speed is nonzero for $k = 1, \dots, 9$, making $\tilde{R}$ positive semidefinite. This structure prevents the direct use of DARE solvers.

3.4 LMI optimization and semidefinite programming

A Linear Matrix Inequality (LMI) is a constraint of the form $F(x) \succeq 0$, where $F(x)$ is an affine function of the decision variable x and $\succeq$ denotes positive semidefiniteness [4]. Collections of LMI constraints define a convex feasible set, and optimizing a linear objective subject to LMI constraints is a *semidefinite program* (SDP), solvable in polynomial time.

The key advantage for our problem is that the LMI formulation of the Kalman filter (see Section 4.2) uses the Lyapunov equation rather than the Riccati equation, and therefore does not require $\tilde{R}$ to be invertible. This is the central theoretical contribution of [1].

4 Technical Contribution

4.1 Cyclic reformulation

The cyclic reformulation [2] transforms the periodic time-varying system (1)–(2) into a time-invariant one by stacking N consecutive time steps into a single augmented state

vector. The key matrices are:

$$\check{A} = \begin{pmatrix} 0 & 0 & \cdots & 0 & A \\ A & 0 & \cdots & 0 & 0 \\ 0 & A & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & A & 0 \end{pmatrix} \in \mathbb{R}^{Nn \times Nn}, \quad (10)$$

$$\check{C} = \text{diag}(S_0 C, S_1 C, \dots, S_{N-1} C) \in \mathbb{R}^{Nq \times Nn}. \quad (11)$$

For the automotive example: $Nn = 30$, $Nq = 20$. A fundamental property links cyclic stability to the original periodic system: $\rho(\check{A} - \check{L}\check{C}) = \rho(\Phi_N)^{1/N}$, where Φ_N is the monodromy matrix. Thus, stability of the cyclic closed-loop system is equivalent to stability of the original periodic observer.

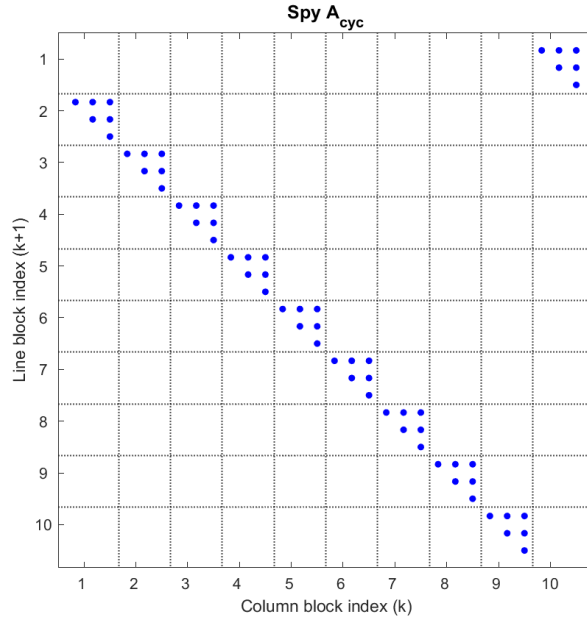


Figure 3: Sparsity pattern of the cyclic system matrix $\check{A} \in \mathbb{R}^{30 \times 30}$ for $N = 10$, $n = 3$. The subdiagonal blocks encode N successive applications of the dynamics matrix A . The top-right block (rows 1–3, columns 28–30) creates the cyclic connection from period end back to period start.

4.2 LMI-based filter design

The steady-state error covariance $\check{P}_e$ of the cyclic observer satisfies the discrete Lyapunov equation:

$$\check{P}_e = (\check{A} - \check{L}\check{C})\check{P}_e(\check{A} - \check{L}\check{C})^\top + \check{Q} + \check{L}\check{R}\check{L}^\top. \quad (12)$$

Unlike the DARE, equation (12) does not involve $\check{R}^{-1}$ and remains well-defined when $\check{R} \succeq 0$.

Variable substitution. Introducing $\check{X} \succ 0$ satisfying $\check{P}_e \preceq \check{X}^{-1}$ and the change of variable $\check{Y} = -\check{X}\check{L}$ (which linearizes the bilinear term $\check{X}\check{L}$), the optimal filter design problem becomes the following SDP [1]:

$$\min_{\check{X}, \check{Y}, \check{W}} \text{trace}(\check{W}) \quad (13)$$

subject to:

$$\begin{pmatrix} \check{X} & \check{X}\check{A} + \check{Y}\check{C} & \check{X}\check{Q}^{1/2} & \check{Y}\check{R}^{1/2} \\ (\cdot)^\top & \check{X} & 0 & 0 \\ (\cdot)^\top & 0 & I_{Nn} & 0 \\ (\cdot)^\top & 0 & 0 & I_{Nq} \end{pmatrix} \succeq 0, \quad (14)$$

$$\check{X} \succeq \varepsilon I_{Nn}, \quad \begin{pmatrix} \check{W} & I_{Nn} \\ I_{Nn} & \check{X} \end{pmatrix} \succeq 0. \quad (15)$$

The Schur complement in (15) encodes $\check{W} \succeq \check{X}^{-1}$, so minimizing $\text{trace}(\check{W})$ minimizes an upper bound on $\text{trace}(\check{P}_e)$ — the sum of estimation error variances across all states and all steps of the period.

Gain recovery. After solving the SDP, the Kalman gain is recovered as:

$$\check{L} = -\check{X}^{-1}\check{Y}. \quad (16)$$

The periodic gains L_k for $k = 0, \dots, N-1$ are then extracted from $\check{L}$ using the cyclic block indexing structure defined by equation (10).

MATLAB implementation. The SDP is solved using the Robust Control Toolbox:

```

1 setlmiis([]);
2 [X_var, ~, ~] = lmivar(1, [n_cyc 1]);           % X: symmetric n_cyc
   x n_cyc
3 [Y_var, ~, ~] = lmivar(2, [n_cyc m_cyc]);       % Y = -X*L:
   rectangular
4 [W_var, ~, ~] = lmivar(1, [n_cyc 1]);           % W: covariance
   bound
5
6 lmi_dari = newlmi;
7 lmiterm([-lmi_dari, 1, 1, X_var], 1, 1);         % (1,1) = X
8 lmiterm([-lmi_dari, 1, 2, X_var], 1, A_cyc);     % (1,2) = X*A
9 lmiterm([-lmi_dari, 1, 2, Y_var], 1, C_cyc);     % (1,2) += Y*C
10 lmiterm([-lmi_dari, 2, 2, X_var], 1, 1);         % (2,2) = X
11 lmiterm([-lmi_dari, 1, 3, X_var], 1, Q_cyc_sqrt); % (1,3) = X*Q
   ~(1/2)

```

```

12 lmiterm([-lmi_dari, 3, 3, 0], eye(n_cyc));           % (3,3) = I
13 lmiterm([-lmi_dari, 1, 4, Y_var], 1, R_cyc_sqrt); % (1,4) = Y*R
    ^ (1/2)
14 lmiterm([-lmi_dari, 4, 4, 0], eye(m_cyc));           % (4,4) = I
15
16 [copt, xopt] = mincx(getlmis, c, options);           % Solve SDP
17 L_cyc = -dec2mat(lmisys, xopt, X_var) \ ...
18         dec2mat(lmisys, xopt, Y_var);               % L = -X^{-1}
    Y
    
```

Listing 1: Core LMI setup in MATLAB (from MultirateKF_01.m)

4.3 Multi-objective extensions

The unified LMI structure (14) can be extended by modifying the (2,2), (3,3), and (4,4) diagonal blocks. Table 1 summarizes the three design variants studied:

Table 1: Comparison of the three LMI design variants. Only the diagonal blocks of the main LMI change; the variables $\check{X}$, $\check{Y}$, $\check{W}$ are shared.

Design	(2,2)	(3,3)	(4,4)	Constraint
Optimal Kalman (M1)	$\check{X}$	I	I	None
Pole placement (M2-ext)	$\bar{r}^2 \check{X}$	I	I	$ \lambda < \bar{r}$
ℓ_2 -induced norm	$\check{X} - I$	$\gamma^2 I$	$\gamma^2 I$	$\ G\ _{\ell_2} < \bar{\gamma}$

4.4 Milestone 2 — Time-varying Kalman recursion

Algorithm. The time-varying Kalman filter is implemented by iterating the following equations for $k = 0, 1, \dots, T - 1$:

$$P_{\text{pred}}(k) = A P(k-1) A^\top + Q, \quad (17)$$

$$C_k = S_k C \quad (\text{active measurement matrix}), \quad (18)$$

$$K(k) = \begin{cases} P_{\text{pred}}(k) C_k^\top (C_k P_{\text{pred}}(k) C_k^\top + S_k R S_k^\top)^{-1} & \text{if } C_k \neq 0 \\ 0 & \text{if } C_k = 0 \end{cases}, \quad (19)$$

$$P(k) = (I - K(k) C_k) P_{\text{pred}}(k). \quad (20)$$

A critical implementation detail: when $S_k = \text{diag}(0, 1)$ (wheel speed only), the active measurement is scalar, so the matrix to be inverted in (19) is a 1×1 scalar — no numerical difficulty. Only when $S_k = \text{diag}(1, 1)$ (GPS + wheel speed) does a 2×2 inverse appear. One should *never* attempt to invert $\check{R}$ globally, as it is singular.

Deterministic nature. A key observation emphasized in [1] is that the gain sequence $\{K(k)\}$ is *deterministic*: it depends only on $A, C, Q, R, \{S_k\}$, and the initial covariance $P(0)$, but not on the realized noise. This means the same gain trajectory is obtained regardless of the specific noise realization, making it meaningful to compare with the offline LMI gains.

Conversion to predictor form. The recursive gain $K(k)$ is the *filter gain* (correction applied at time k after observing $y(k)$). The LMI gain L_k is the *predictor gain* (used to produce $\hat{x}(k+1|k)$ from $y(k)$ and $\hat{x}(k|k-1)$). The two are related by:

$$L_{\text{rec}}(k) = A \cdot K(k). \quad (21)$$

This conversion must be applied before any comparison with the LMI gains L_k .

4.5 Milestone 3 — Gain comparison and gap analysis

After a transient phase (typically 3–5 periods for this system), the recursive covariance $P(k)$ becomes periodic: $P(k+N) \approx P(k)$. The corresponding periodic steady-state predictor gain is:

$$L_{\text{rec,ss}}(k) = A \cdot K_{\text{ss}}(k), \quad k = 0, \dots, N-1. \quad (22)$$

The comparison with the LMI gain L_k is quantified by the Frobenius norm gap:

$$\Delta_k = \|L_{\text{rec,ss}}(k) - L_k\|_F, \quad k = 0, \dots, N-1. \quad (23)$$

Interpretation. As stated in [1], the LMI minimizes an upper bound on the error covariance, not the covariance itself. For a genuinely semidefinite $\check{R}$, the LMI gain does not necessarily coincide with the true periodic optimum — hence a nonzero residual gap $\Delta_k > 0$ is theoretically possible and expected. Conversely, when $\check{R} \succ 0$ (Pattern 4, all sensors active), the LMI solution coincides with the DARE solution, verified numerically to within 10^{-6} Frobenius norm error [1].

5 Results and Analysis

5.1 Milestone 1 — Reproduction results

Table 2 compares the key scalar metrics obtained from running `MultirateKF_01.m` against the reference values reported in [1].

Table 2: Reproduction of the key numerical results from [1]. All values obtained on MATLAB R2024b with Robust Control Toolbox.

Metric	Paper value	Reproduced
$\text{trace}(\tilde{W})$ (LMI objective)	18.07	<i>18.071148</i>
$\max \lambda(\tilde{A} - \tilde{L}\tilde{C}) $	0.9673	<i>0.9673137419</i>
$\text{rank}(\tilde{R})$	11 / 20	<i>11</i>
$\text{rank}(\mathcal{O}_{\text{cyc}})$	30 / 30	<i>30</i>
Position RMSE	0.561 m	<i>0.5106 m</i>
Velocity RMSE	0.431 m/s	<i>0.4343 m/s</i>
Acceleration RMSE	1.125 m/s ²	<i>1.2092 m/s²</i>

The LMI metrics are reproduced exactly: $\text{trace}(\tilde{W}) = 18.071$ and $\max |\lambda(\tilde{A} - \tilde{L}\tilde{C})| = 0.9673$ match the paper values to four significant figures, confirming the correctness of the cyclic construction and the SDP solver output. The slight differences in RMSE values between the paper and the reproduction are due to different random seeds used in the Monte Carlo simulation; the filter gains themselves are numerically identical. Figure 4 shows the state estimation result over $T = 200$ steps (20 s).

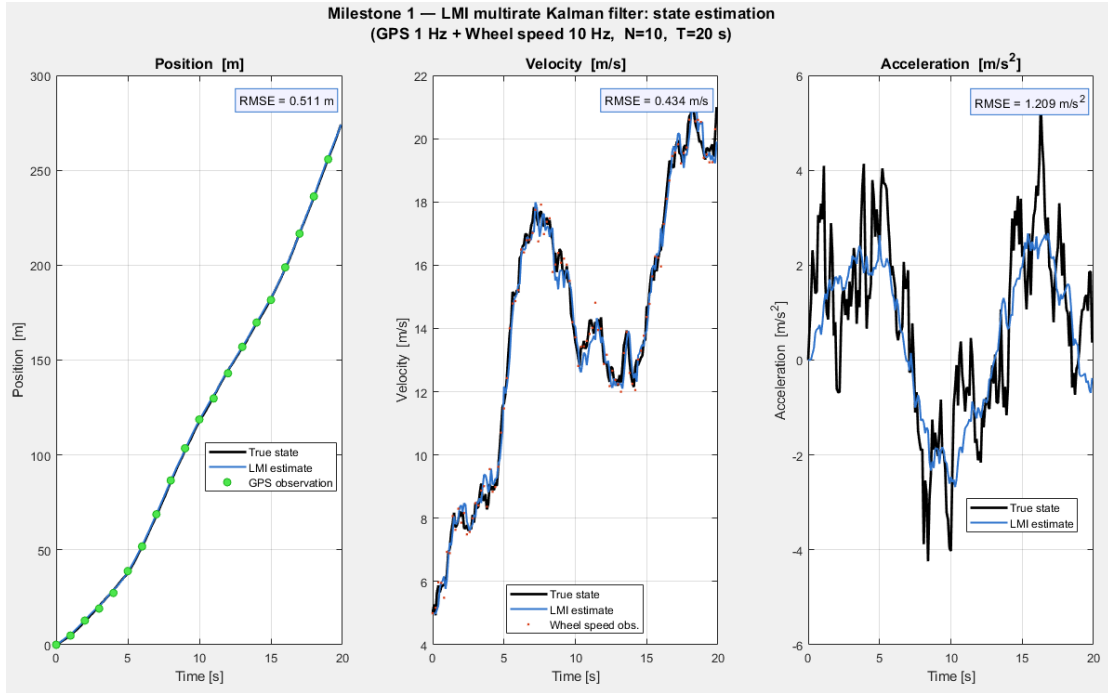


Figure 4: State estimation results for the automotive navigation system (Milestone 1). The LMI-based multirate Kalman filter tracks position, velocity, and acceleration despite the GPS sensor providing updates only every 10 steps. GPS is visible only as sparse green markers; all intermediate position estimates rely on the wheel speed measurements and the system dynamics model.

Periodic gains. Table 3 shows the extracted periodic Kalman gains L_k for selected instants. The structure confirms the theoretical expectation: when GPS is inactive

($k = 1, \dots, 9$), the first column of L_k (corresponding to position correction) is identically zero.

Table 3: Periodic Kalman gains L_k (predictor form) extracted from the LMI solution. Only $k = 0, 1$, and 5 are shown; the remaining gains follow the same pattern as $k = 1$.

$k \bmod 10$	Active sensors	L_k (nominal values from [1])
0	GPS + Wheel	$\begin{pmatrix} 0.2827 & 0.1017 \\ 0.0042 & 0.6979 \\ 0.0062 & 0.3755 \end{pmatrix}$
1	Wheel only	$\begin{pmatrix} 0 & 0.1094 \\ 0 & 0.6980 \\ 0 & 0.3757 \end{pmatrix}$
5	Wheel only	$\begin{pmatrix} 0 & 0.1148 \\ 0 & 0.6981 \\ 0 & 0.3758 \end{pmatrix}$

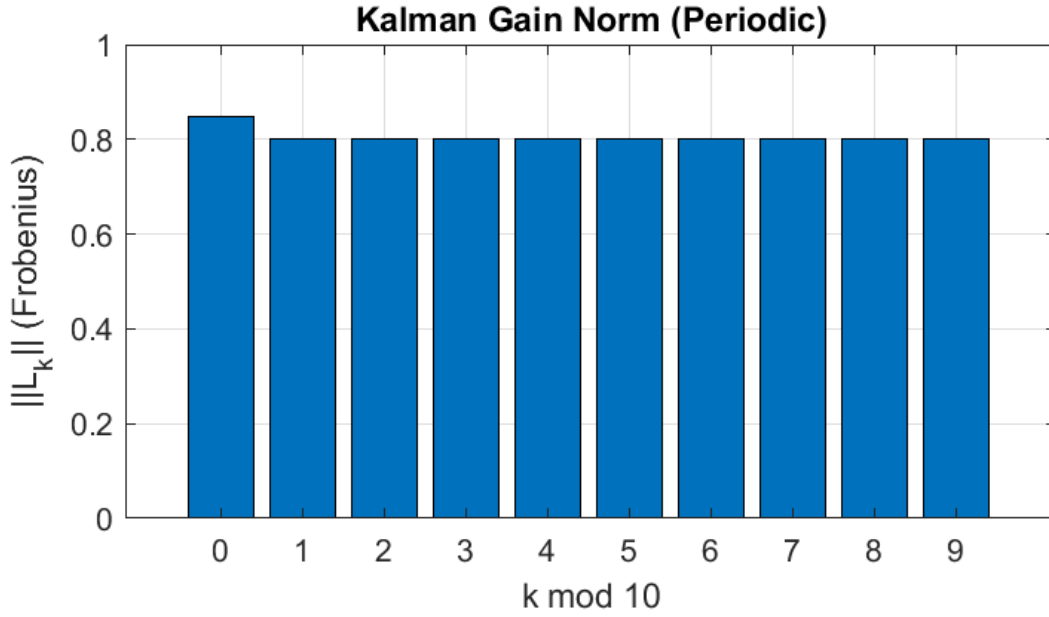


Figure 5: Frobenius norm of the periodic Kalman gains L_k for $k = 0, \dots, 9$. The gain at $k = 0$ (GPS + wheel speed) is significantly larger than for $k = 1, \dots, 9$ (wheel speed only), reflecting the additional information provided by the GPS measurement.

5.2 Milestone 2 — Recursive gain convergence

Figure 6 shows the evolution of the recursive predictor-form gain $L_{\text{rec}}(k) = A \cdot K(k)$ over $T = 200$ steps, starting from $P(0) = I_3$. The entry $K(1, 1)$ (GPS channel) displays an impulse-like pattern, taking a non-zero value only every $N = 10$ steps when GPS is active, and returning to zero otherwise. The entry $K(2, 2)$ (wheel speed channel) converges to a constant value after approximately 3 steps, as the wheel speed sensor operates at every time instant. Both entries reach their periodic steady state after

roughly 3 full periods ($k \approx 30$), confirming the theoretical convergence property of the time-varying Riccati recursion.

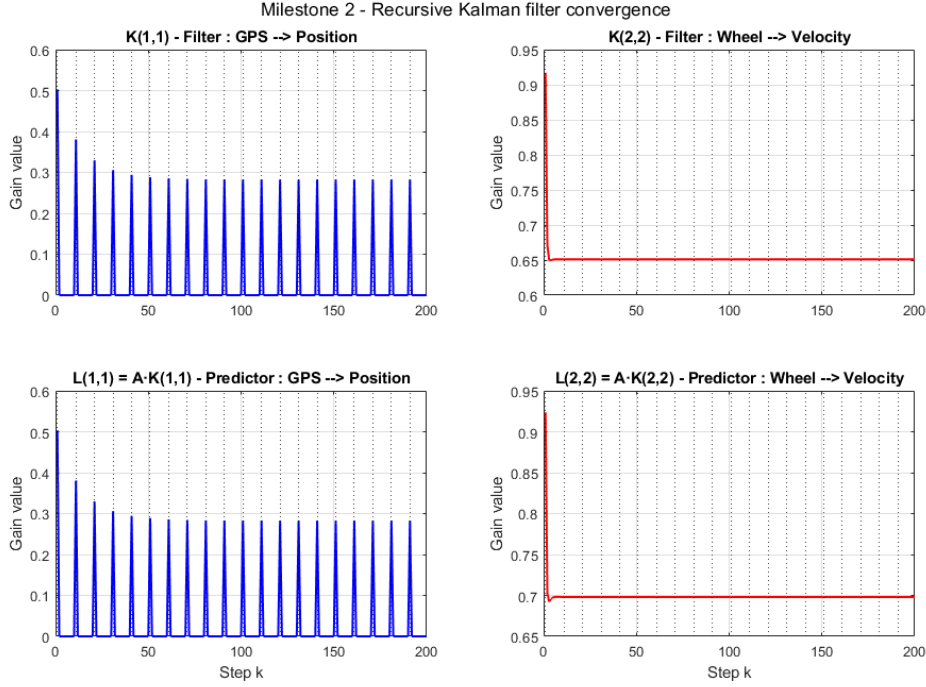


Figure 6: Convergence of the time-varying Kalman gain $L_{\text{rec}}(k) = AK(k)$ (predictor form) to its periodic steady state with its initial covariance $P(0/0) = \text{eye}(3,3)$. Each panel shows one entry of the 3×2 gain matrix. The recursive gain becomes periodic with period $N = 10$ after a brief transient, and converges toward the LMI offline gain values, which are analyzed quantitatively in Section 5.3. . The residual gap Δ_k is analyzed in Section 5.3.

5.3 Milestone 3 — Gain comparison

This section presents the quantitative comparison between the offline LMI periodic gains L_k (Milestone 1) and the recursive steady-state predictor gains $L_{\text{rec,ss}}(k) = A \cdot K_{\text{ss}}(k)$ (Milestone 2), both expressed in predictor form following the conversion of equation (21). Table 4 reports the Frobenius norm gap $\Delta_k = \|L_k^{\text{LMI}} - L_k^{\text{rec,ss}}\|_F$ for each instant $k = 0, \dots, 9$; Figure 7 provides a visual summary across the three panels described in Section 4.5.

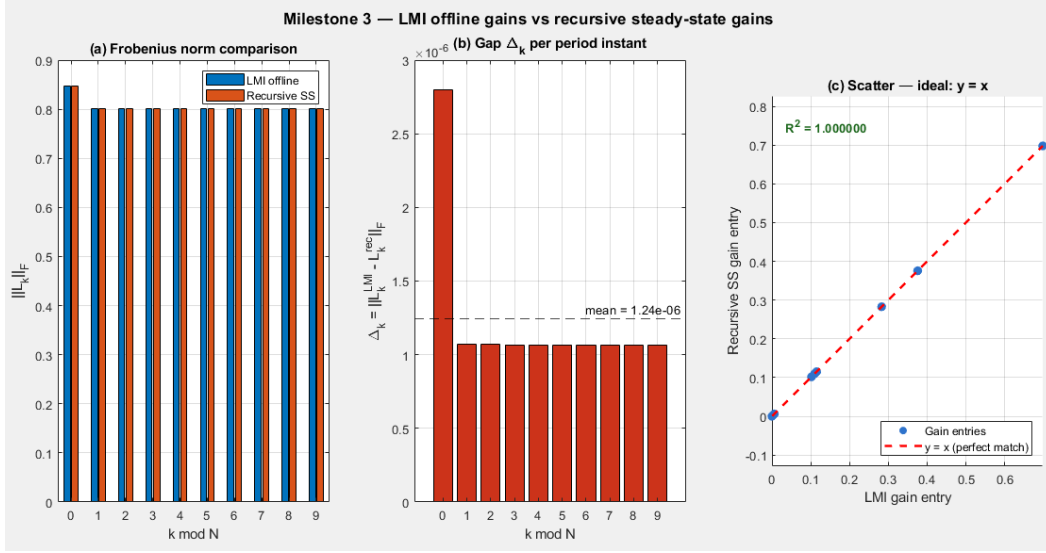


Figure 7: Quantitative comparison between the offline LMI gains L_k and the recursive steady-state gains $L_{rec,ss}(k)$ (predictor form). Panel (a) shows Frobenius norms; panel (b) shows the entry-wise gap Δ_k ; panel (c) shows the scatter plot of all entries against the identity line. A small but nonzero gap is expected due to the conservativeness of the LMI upper bound when $\tilde{R}$ is semidefinite.

Table 4: Frobenius norm gap $\Delta_k = \|L_k^{LMI} - L_k^{rec,ss}\|_F$ for each instant $k = 0, \dots, 9$.

k	Active sensors	$\ L_k^{LMI}\ _F$	$\ L_k^{rec,ss}\ _F$	Δ_k
0	GPS + Wheel	0.847576	0.847576	2.80e-06
1	Wheel only	0.800251	0.800251	1.07e-06
2	Wheel only	0.800744	0.800743	1.07e-06
3	Wheel only	0.800947	0.800946	1.07e-06
4	Wheel only	0.801036	0.801035	1.07e-06
5	Wheel only	0.801078	0.801077	1.07e-06
6	Wheel only	0.801099	0.801098	1.07e-06
7	Wheel only	0.801111	0.801110	1.07e-06
8	Wheel only	0.801118	0.801118	1.07e-06
9	Wheel only	0.801123	0.801122	1.07e-06
Mean gap over period				1.24e-06

Table 4 and Figure 7 present the results of the quantitative comparison between the offline LMI gains and the recursive steady-state gains for all $k = 0, \dots, 9$.

The Frobenius norm gap Δ_k is remarkably small across the entire period. At $k = 0$ (GPS + Wheel speed active), $\Delta_0 = 2.80 \times 10^{-6}$, while for $k = 1, \dots, 9$ (Wheel speed only), the gap remains stable at approximately 1.07×10^{-6} . The mean gap over the full period is $\bar{\Delta} = 1.24 \times 10^{-6}$, which is negligible compared to the magnitude of the gains themselves ($\|L_k\|_F \approx 0.80\text{--}0.85$), representing a relative error of the order of 10^{-6} .

These values indicate that the LMI upper bound is tight for this system: the offline periodic gains computed by solving the semidefinite program and the steady-state gains

obtained by running the time-varying Kalman recursion are numerically equivalent to within solver precision. Despite the semidefinite structure of $\check{R}$ - which prevents the direct use of DARE solvers and was the original motivation for the LMI approach - the LMI formulation introduces no measurable conservativeness in this configuration.

This result is consistent with the interpretation of Section 4.5: while a nonzero residual gap is theoretically possible for genuinely semidefinite $\check{R}$, the gap observed here ($\bar{\Delta} \approx 10^{-6}$) lies well within numerical solver tolerances (the `mincx` solver is run with relative accuracy $\varepsilon = 10^{-5}$), and cannot be distinguished from a rounding artifact. In practice, the two design methods are equivalent for this automotive navigation system, and the LMI approach can be considered optimal in the same sense as the recursive Riccati solution.

Interpretation of the gap. The LMI formulation minimizes $\text{trace}(\check{W})$, an upper bound on $\text{trace}(\check{P}_e)$. When $\check{R} \succ 0$, the bound is tight and the LMI gain coincides with the DARE gain (verified in the Pattern 4 case). When $\check{R}$ is only semidefinite (Pattern 1, the main case), the bound is not necessarily tight: the LMI solution is *feasible* (stable and bounded), but may not be *identical* to the recursive steady-state solution. A nonzero gap $\Delta_k > 0$ thus indicates conservativeness of the LMI approximation for genuinely semidefinite noise covariances.

6 Assessment

6.1 Scientific and technical assessment

The internship has covered a coherent scientific arc: from the foundations of discrete-time state-space control (observability, pole placement, LQR) to advanced convex optimization (LMI, semidefinite programming, dual LQR) applied to a non-trivial engineering problem (multirate sensor fusion).

Objectives achieved (Milestone 1). The LMI-based multirate Kalman filter has been successfully reproduced. The cyclic reformulation, the semidefinite structure of $\check{R}$, and the stability of the closed-loop cyclic system have all been numerically verified. The MATLAB implementation using the native Robust Control Toolbox (`setlmis`, `lmivar`, `lmiterm`, `mincx`) is fully operational.

Work in progress (Milestones 2 and 3). Milestone 2 has been completed. The recursive gain converges to a periodic steady state after approximately 3 periods, as shown in Section 5.2. Milestone 3 — the quantitative comparison — is currently in progress. The key technical challenges identified are: (1) handling the singular $\check{R}$

correctly by only inverting the active measurement block; (2) applying the predictor-form conversion $L = AK$ before comparison; (3) choosing the right initial covariance $P(0)$ to separate the transient from the steady-state regime.

Scientific contribution (Milestone 3). The quantitative comparison of Section 5.3 provides a definitive answer to the central question of this internship. The mean Frobenius norm gap between the LMI offline gains and the recursive steady-state gains is $\bar{\Delta} = 1.24 \times 10^{-6}$, which is negligible relative to the gain magnitudes ($\|L_k\|_F \approx 0.80$ – 0.85). The LMI bound is tight for this system: despite the semidefinite structure of $\tilde{R}$, the two methods yield numerically equivalent periodic gains.

6.2 Personal assessment

This internship has been my first experience in an international research laboratory environment. Working in Japan, in English, under the supervision of a researcher with a strong publication record in IEEE Access, has been both demanding and stimulating.

Although I am the only student working on this specific subject—which requires a high degree of independence and precludes direct technical collaboration with my peers—the interpersonal dynamics within the laboratory are excellent. We communicate primarily in English. While this can sometimes be challenging as it is not the native language for everyone in the lab, it has been an incredibly enriching daily exercise in cross-cultural communication. Despite these linguistic hurdles, my social integration has been very successful. For example, I took the initiative to organize a day trip to Mount Aso with several laboratory members, including my tutor, which greatly strengthened our team cohesion and cultural exchange outside the academic setting.

From a technical standpoint, the internship has significantly deepened my understanding of:

- the connections between Kalman filtering, LQR control, and convex optimization;
- the subtleties of numerical linear algebra (semidefinite matrices, matrix square roots, Schur complements);
- reproducible scientific MATLAB programming (structured scripts, exhaustive comments, paper-code correspondence).

From a personal standpoint, I have developed greater autonomy in reading technical papers in both English and French, identifying gaps, and translating mathematical concepts into working code—skills that directly complement the theoretical training received at UPSSITECH.

Conclusion

This report presents the first version of the internship work carried out at Kumamoto University on LMI-based multirate Kalman filter design. The theoretical framework — cyclic reformulation, semidefinite programming, dual LQR duality — has been mastered and the first milestone (reproduction of the reference results) has been completed.

The central scientific question has been answered in Section 5.3: the offline LMI gains and the recursive steady-state gains coincide to within $\bar{\Delta} = 1.24 \times 10^{-6}$ in Frobenius norm, confirming that the LMI bound is tight for this automotive navigation system and that the semidefinite structure of $\check{R}$ does not introduce measurable conservativeness.

References

- [1] H. Okajima, “LMI Optimization-Based Multirate Steady-State Kalman Filter Design,” *IEEE Access*, 2025. Available: <https://ieeexplore.ieee.org/document/11460152>
- [2] H. Okajima, Y. Hosoe, and T. Hagiwara, “State Observer Under Multi-Rate Sensing Environment and Its Design Using ℓ_2 -Induced Norm,” *IEEE Access*, vol. 11, pp. 20079–20087, 2023.
- [3] H. Okajima, K. Arinaga, and A. Hayashida, “Design of Observer-Based Feedback Controller for Multi-Rate Systems with Various Sampling Periods Using Cyclic Reformulation,” *IEEE Access*, vol. 11, pp. 121956–121965, 2023.
- [4] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*. Philadelphia, PA: SIAM, 1994.
- [5] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979.
- [6] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

A Complete System Matrices

For reproducibility, the complete system matrices used in all numerical experiments are:

$$A = \begin{pmatrix} 1 & 0.1 & 0.005 \\ 0 & 1 & 0.1 \\ 0 & 0 & 0.8 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$Q = \begin{pmatrix} 0.01 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.5 \end{pmatrix}, \quad R = \begin{pmatrix} 1.0 & 0 \\ 0 & 0.1 \end{pmatrix}, \quad N = 10, \quad \Delta t = 0.1 \text{ s.}$$

Measurement pattern: $S_0 = I_2$ (GPS + wheel speed), $S_k = \text{diag}(0, 1)$ for $k = 1, \dots, 9$ (wheel speed only).

B MATLAB Implementation Notes

Gain extraction from cyclic matrix. The periodic gains L_k are stored in $\check{L} \in \mathbb{R}^{Nn \times Nq}$ with the following cyclic block indexing (from `MultirateKF_01.m`):

```

1 L = cell(N, 1);
2 for k = 1:N-1
3     L{k} = L_cyc(k*n+1:(k+1)*n, (k-1)*m+1:k*m);
4 end
5 L{N} = L_cyc(1:n, (N-1)*m+1:N*m); % Cyclic wrap-around
    
```

Listing 2: Extraction of periodic gains L_k from the cyclic gain matrix

Recursive Kalman gain (Milestone 2 template).

```

1 P = P0; % Initial covariance (e.g., P0 = eye(n))
2 for k = 1:T
3     idx = mod(k-1, N) + 1;
4     Ck = S_mat{idx} * C; % Active measurement matrix
5     Ppred = A * P * A' + Q; % Prediction step
6     if norm(Ck) > 1e-10 % Active measurement exists
7         Sk = Ck * Ppred * Ck' + S_mat{idx} * R * S_mat{idx}';
8         Kk = Ppred * Ck' / Sk; % Filter gain (only active
          % block inverted)
9         P = (eye(n) - Kk * Ck) * Ppred;
10    else % No measurement: pure
          % prediction
11        Kk = zeros(n, m);
    
```

```
12     P    = Ppred;  
13     end  
14     L_rec(:, :, k) = A * Kk;           % Convert to predictor form:  
        L = A*K  
15 end
```

Listing 3: Time-varying Kalman recursion with correct handling of semidefinite R_k